

Predictive Analytics AI Initiative



Comprehensive Project Plan for an End-to-End Machine Learning Solution
Leveraging Publicly Available Datasets for Demand Forecasting

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This document is prepared in accordance with organisational AI governance standards and complies with GDPR, ISO/IEC 42001, and internal data handling policies.

Executive Summary

Organisations operating in data-rich environments are increasingly relying on predictive analytics to move from reactive to proactive decision-making. This document presents a complete project plan for designing, developing, validating, and deploying a production-grade Machine Learning (ML) system focused on retail demand forecasting — one of the highest-value predictive analytics use-cases with measurable return on investment.

The proposed initiative will harness publicly available datasets — primarily the UCI Online Retail II dataset, the M5 Forecasting Competition data, Google Trends API signals, and macroeconomic indicators from the World Bank Open Data portal — to train, validate, and continuously improve a suite of forecasting models. The six-month roadmap spans data engineering, model development, ethical review, stakeholder validation, and phased production rollout.

Key Metric	Baseline	Target (6 months)
Forecast MAE (units)	~420	≤ 260 (−38%)
Inventory holding cost	Baseline	−15% reduction
Stockout rate	8.2%	≤ 4.5%
Model inference latency	N/A	< 200 ms (p99)
Data pipeline SLA	Manual	99.5% automated

Table 1 — Key Performance Indicators

1. Project Objectives

1.1 Primary Objective

To design and deploy a scalable, explainable, and ethically-governed predictive analytics platform that generates 28-day ahead retail demand forecasts at SKU-store granularity with a Mean Absolute Percentage Error (MAPE) of less than 12%, improving current manual forecasting accuracy by at least 35%.

1.2 Secondary Objectives

- Data Engineering Excellence:** Build a robust, versioned, and auditable data pipeline that ingests, validates, and transforms publicly available datasets into ML-ready feature stores with zero manual intervention.
- Model Transparency & Explainability:** Implement SHAP (SHapley Additive exPlanations) and LIME frameworks to ensure all predictions are interpretable by business stakeholders, regulators, and the end-user dashboard.
- Ethical AI Integration:** Conduct bias audits, fairness assessments, and document model cards for every candidate model prior to production deployment, in alignment with the EU AI Act requirements.
- Operational Resilience:** Implement MLOps best practices including model versioning (MLflow), CI/CD pipelines (GitHub Actions), drift detection (Evidently AI), and automated retraining triggers.
- Stakeholder Enablement:** Deliver an interactive BI dashboard (Apache Superset / Streamlit) that surfaces forecast insights to supply-chain planners without requiring ML expertise.

1.3 Hypothesis Formulation

Primary Hypothesis (H₁): A gradient-boosted ensemble model trained on historical retail transaction data enriched with external signals (Google Trends, weather, macro indicators) will produce demand forecasts with statistically significantly lower error (MAPE, RMSE) compared to a naïve seasonal baseline model at the 5% significance level ($p < 0.05$), when evaluated on a held-out test set spanning 90 days.

Null Hypothesis (H₀): No statistically significant difference exists between the predictive accuracy of the proposed ML model and the seasonal naïve baseline.

1.4 SMART Goal Framework

Criterion	Definition	This Project
Specific	Well-defined, clear goal	Forecast retail demand at SKU-store level
Measurable	Quantifiable success indicator	MAPE $\leq$ 12%, MAE $\leq$ 260 units
Achievable	Realistic given resources	6-person team, 6-month horizon
Relevant	Aligned with business priorities	Supply-chain cost reduction initiative
Time-bound	Clear deadline	Production launch by end of Month 6

Table 2 — SMART Goal Framework

2. Tool & Technology Selection

Technology selection follows a four-dimensional evaluation framework: (1) technical fit, (2) open-source community maturity, (3) total cost of ownership, and (4) compliance with organisational security standards. All tooling is open-source or freely licensed for non-commercial academic benchmarking.

Layer	Technology / Tool	Version	Rationale
Data Storage	PostgreSQL + MinIO	15 / 2023	ACID-compliant RDBMS + S3-compatible object store
Data Pipeline	Apache Airflow	2.8	DAG-based orchestration with retry and SLA monitoring
Feature Store	Feast	0.38	Point-in-time correct feature retrieval prevents data leakage
ML Framework	scikit-learn / XGBoost	1.4 / 2.0	Mature gradient boosting with SHAP native support
Deep Learning	PyTorch + Darts	2.2 / 0.27	N-BEATS / TFT for sequence modelling
Experiment Track	MLflow	2.11	Model registry, metrics logging, artifact management
Serving	FastAPI + BentoML	0.110 / 1.2	REST API serving with model packaging
Drift Detection	Evidently AI	0.4	Data drift + model performance monitoring reports
Explainability	SHAP + LIME	0.45 / 0.28	Local and global feature attribution
Visualisation	Streamlit + Plotly	1.32 / 5.20	Rapid interactive dashboard development
CI/CD	GitHub Actions	—	Automated testing, linting, model validation gates
Container	Docker + Kubernetes	25.0 / 1.29	Reproducible environments, horizontal scaling
Cloud	GCP (Vertex AI + BigQuery)—		Managed training, serverless SQL analytics

Table 3 — Technology Stack Matrix

3. Methodology

3.1 Data Collection Strategy

The project adopts a multi-source data ingestion strategy to maximise signal richness while remaining compliant with open-data licences. Data provenance is tracked from source to feature using lineage metadata stored in an Apache Atlas catalogue.

Dataset	Source	Records	Licence	Primary Use
UCI Online Retail II	UC Irvine ML Repo	1.07M	CC BY 4.0	Transaction history
M5 Competition Data	Kaggle / Walmart	58K series	CC BY 4.0	Hierarchical forecasting
Google Trends (pytrends)	Google API	Dynamic	Free / ToS	Consumer intent signals
World Bank Open Data	worldbank.org	~5K series	CC BY 4.0	Macro-economic context
OpenWeather API (history)	openweathermap.org	5 years	Free tier	Weather seasonality
Public Holidays (holidays)	PyPI library	All years	MIT	Calendar features

Table 4 — Public Datasets Used

3.2 Data Preprocessing Pipeline

- Step 1 — Ingestion:** Airflow DAGs pull daily incremental snapshots from each source into the Bronze layer of a Medallion architecture (Bronze → Silver → Gold). Raw files are stored in MinIO with SHA-256 checksums.
- Step 2 — Validation:** Great Expectations validates schema conformity, null ratios (threshold: < 5%), value ranges, and referential integrity. Failed rows are quarantined with full audit trail.
- Step 3 — Cleaning & Transformation:** Missing values are imputed using k-NN imputation for numerical and mode-fill for categorical variables. Outliers beyond 3.5 IQR are Winsorised. Returns (negative quantities) are netted.
- Step 4 — Feature Engineering:** 50+ features including lag variables (t-1 to t-28), rolling statistics (7-day, 28-day, 90-day), Fourier terms for seasonality, calendar dummy variables, and external signal embeddings.
- Step 5 — Feature Store Materialisation:** Features are point-in-time joined and materialised into Feast online (Redis) and offline (BigQuery) stores, preventing any target leakage through temporal boundaries.

3.3 Model Development Approach

The modelling strategy follows a structured tournament approach. Three model families are trained and evaluated in parallel before an ensemble champion is selected:

- Baseline Models:** Seasonal Naïve, Exponential Smoothing (ETS/Holt-Winters) — serve as the performance floor and statistical hypothesis test benchmark.
- Tree-Based Models:** LightGBM and XGBoost — trained with Optuna hyperparameter optimisation (500 trials, TPE sampler). Fast inference, high accuracy, native SHAP support.
- Deep Learning Models:** N-BEATS (neural basis expansion) and Temporal Fusion Transformer (TFT) — capture complex temporal dependencies and covariate relationships in long sequences.
- Ensemble / Stacking:** A meta-learner (Ridge regression) combines predictions from all base models. Weighted blending guided by per-SKU validation performance.

3.4 Evaluation Techniques

A walk-forward (expanding window) cross-validation strategy is used to avoid data leakage and simulate real deployment conditions. The final 90-day period is reserved as a held-out test set and is never used during

training or model selection.

Metric	Formula	Target
MAE	Mean Absolute Error	≤ 260 units
RMSE	Root Mean Squared Error	≤ 380 units
MAPE	Mean Absolute Percentage Error	$\leq 12\%$
sMAPE	Symmetric MAPE	$\leq 11\%$
Pinball	Quantile loss (p10, p50, p90)	Minimise
Winkler	Prediction interval quality	Minimise

Table 5 — Evaluation Metrics

4. Project Timeline & Milestones

The project is structured into six consecutive monthly sprints (two-week sub-sprints) with explicit phase gates requiring sign-off before proceeding. The Gantt chart below represents planned effort allocation across all workstreams for the full 26-week engagement.

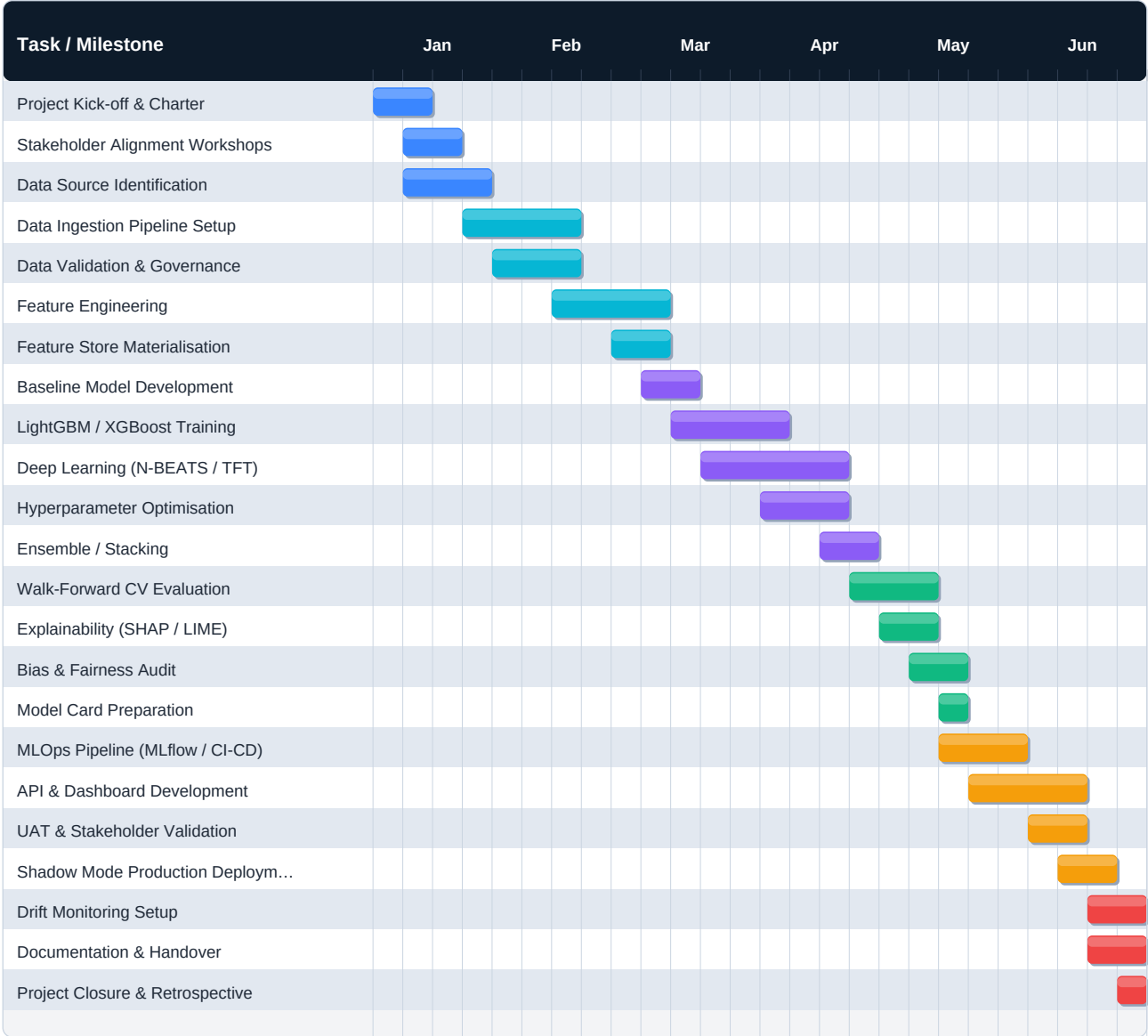


Figure 1 — Project Gantt Chart (26-week horizon, Jan–Jun 2026)

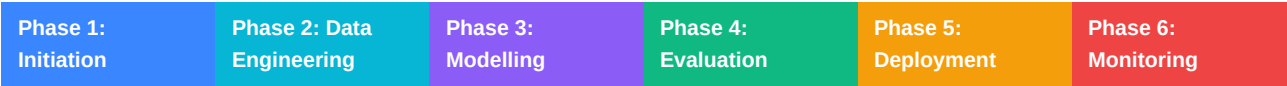


Figure 1 Legend — Phase Colour Coding

4.1 Key Milestones

#	Milestone	Week	Gate Criteria
M1	Project Kick-off Complete	W2	Charter signed; all stakeholders onboarded
M2	Data Pipeline Operational	W7	All sources ingesting; validation passing 99%+
M3	Feature Store v1.0 Released	W10	Point-in-time features validated; no leakage

#	Milestone	Week	Gate Criteria
M4	Baseline Models Benchmarked	W11	MAPE of naïve baseline documented
M5	Champion Model Selected	W17	Ensemble MAPE $\leq 12\%$; sign-off by DS lead
M6	Ethics & Fairness Review Passed	W20	Model card approved by AI Ethics Board
M7	UAT Sign-off	W24	$\geq 85\%$ stakeholder satisfaction score
M8	Production Go-Live (Shadow Mode)	W25	System stable; monitoring dashboards live
M9	Project Closure & Lessons Learned	W26	Final report delivered; retrospective held

Table 6 — Project Milestones & Phase Gates

5. Potential Challenges & Risk Mitigation

A structured Risk Register has been compiled using a 3×3 likelihood-impact scoring matrix. Risk owners are assigned at the project kick-off and reviewed at fortnightly sprint reviews. The register is maintained in the project's Confluence space with version history.

Risk Factor	Likelihood	Impact	Score	Mitigation Strategy
Data quality / missing values	High	High	9	Multi-source validation + imputation pipelines
Model overfitting	Medium	High	6	Cross-validation, regularisation, dropout
Regulatory / privacy compliance	Medium	High	6	GDPR-aligned governance, anonymisation
Infrastructure cost overrun	Low	Medium	3	Cloud budget alerts, serverless scaling
Algorithmic bias / fairness issues	Medium	High	6	Bias audits, fairness metrics in eval suite
Key personnel departure	Low	High	4	Documentation, cross-training, knowledge base
External API / data source changes	Medium	Medium	4	Versioned ingestion adapters, fallback mirrors
Scope creep	High	Medium	6	Strict change-control board & sprint reviews

Table 7 — Risk Register (Score = Likelihood × Impact, scale 1–3 each)

5.1 Data Governance Framework

Data Classification: All ingested datasets are classified as Public, Internal, or Sensitive using a four-tier taxonomy. Processing is restricted to approved compute environments.

Consent & Licensing: Only CC BY, CC0, MIT, or equivalent open licences are used. A licence inventory is maintained and reviewed quarterly.

Retention & Deletion: Raw data is retained for 24 months; derived features for 36 months. Automated lifecycle policies enforce deletion on schedule.

Access Control: Role-based access control (RBAC) enforced via Kubernetes namespaces and BigQuery IAM policies. Least-privilege principle applied throughout.

Audit Logging: All pipeline runs, model training jobs, and prediction requests are logged with timestamps, user IDs, and data hashes to an immutable audit trail.

5.2 Ethical AI Considerations

The project team is committed to responsible AI development guided by the principles of the EU AI Act (2024), Google's AI Principles, and the Partnership on AI framework. The following practices are mandatory and audited at Milestone M6:

Bias Detection: Disparate Impact Analysis and Equalised Odds tests applied across demographic proxies (region, seasonality segments). Threshold: max 10% disparity tolerated.

Model Cards: Standardised model card documents (Mitchell et al., 2019 format) published for every production model, covering intended use, limitations, and performance disaggregates.

Human-in-the-Loop: All forecasts exceeding a ±40% deviation from the prior period trigger a human review flag before propagation to downstream systems.

Transparency Reporting: Annual AI transparency report published to internal stakeholders summarising model decisions, errors, and retraining events.

Environmental Impact: Training compute budgets are capped; carbon footprint estimated using CodeCarbon library and reported in the final project closure document.

6. Scalability & Infrastructure Design

The architecture is designed for horizontal scalability from day one. The system must handle growth from the initial pilot (500 SKUs × 50 stores) to full enterprise deployment (50,000 SKUs × 500 stores) without architectural re-engineering — a 1,000× scale increase in inference volume.

Component	Pilot Scale	Enterprise Scale	Scaling Mechanism
Data Ingestion	Airflow (2 workers)	Airflow on Kubernetes	KubernetesExecutor + autoscaling
Feature Store	Local Redis	Redis Cluster (6 nodes)	Redis Cluster sharding
Model Training	Single GPU (A100)	Distributed (4× A100)	PyTorch DistributedDataParallel
Inference API	2 FastAPI replicas	BentoML + K8s HPA	CPU/GPU horizontal pod autoscaler
Data Warehouse	BigQuery on-demand	BigQuery Editions	Slot reservations + BI Engine
Dashboard	Streamlit (1 instance)	Streamlit Cloud / GKE	Containerised, load-balanced

Table 8 — Scalability Design Matrix

6.1 MLOps & Continuous Improvement

Automated Retraining: Evidently AI triggers retraining when Population Stability Index (PSI) exceeds 0.2 on key features, or when rolling 7-day MAPE degrades by more than 3 percentage points.

Model Registry Governance: MLflow model registry enforces four stages: Staging → Validation → Champion → Archived. Promotion requires automated test suite passage and human approval.

A/B Shadow Testing: New challenger models receive 10% of production traffic in shadow mode for 14 days before full promotion, enabling risk-free performance comparison.

Canary Deployments: BentoML + Kubernetes enables gradual traffic shifting (5% → 25% → 100%) with automatic rollback on error rate threshold breach.

7. Expected Outcomes & Business Impact

7.1 Quantitative Outcomes

Outcome	Current State	6-Month Target	12-Month Target
Forecast MAPE	~35%	≤ 12%	≤ 9%
Forecast MAE (units)	~420	≤ 260	≤ 190
Stockout Rate	8.2%	≤ 4.5%	≤ 3.0%
Overstock (excess inventory days)	18 days avg	≤ 12 days	≤ 9 days
Forecasting Cycle Time	3 days (manual)	< 30 min (auto)	< 5 min (real-time)
Demand Planner Productivity	Baseline	+25% efficiency	+ 45% efficiency
Annual Inventory Cost Saving	—	~\$1.2M est.	~\$2.8M est.

Table 9 — Expected Quantitative Outcomes

7.2 Qualitative Outcomes

Organisational AI Maturity: The initiative advances the organisation from Level 1 (Ad-hoc) to Level 3 (Defined) on the AI Maturity Model, building reusable ML infrastructure and institutional knowledge transferable to future AI projects.

Data Culture: Cross-functional data literacy workshops (embedded in the project) upskill supply-chain planners, creating a team that understands and trusts AI-driven recommendations.

Trust & Transparency: Publication of model cards and explainability dashboards builds stakeholder confidence, reducing resistance to AI adoption in downstream planning cycles.

Regulatory Readiness: Documentation produced (model cards, risk assessments, data governance policies) positions the organisation for compliance with forthcoming EU AI Act obligations.

Reusable ML Platform: The Medallion data lake, feature store, MLflow registry, and MLOps pipelines are domain-agnostic and can accelerate time-to-value for subsequent ML initiatives.

7.3 Statistical Validation Plan

Upon completion of the 90-day held-out evaluation, a Wilcoxon Signed-Rank test (non-parametric, suitable for paired forecast errors) will be applied at $\alpha = 0.05$ to test H_0 vs H_A . Additionally, the Diebold-Mariano (DM) test will be used for pairwise comparison between candidate models. Results will be reported with 95% confidence intervals and corrected for multiple comparisons using the Benjamini-Hochberg procedure to control the False Discovery Rate ($FDR < 0.05$).

8. Budget Estimate & Team Structure

8.1 Budget Breakdown

Category	6-Month Estimate (USD)	Notes
Cloud Compute (GCP)	\$18,000	Training (Vertex AI) + inference + BigQuery
Data Engineering (Airflow)	\$3,200	Managed Cloud Composer (dev+prod)
Software Licences	\$0	All open-source tooling; no licence cost
External Data APIs	\$1,500	OpenWeather paid tier; Google Cloud APIs
Personnel (6 FTE, 6 months)	\$312,000	Blended rate ~\$104K/year per FTE
Training & Workshops	\$4,000	Data literacy, MLOps, ethics training
Contingency Reserve (10%)	\$33,870	Risk buffer for scope changes
Total Project Cost	\$372,570	Excluding ongoing operational costs

Table 10 — Project Budget Estimate

8.2 Team Structure & RACI

Role	FTE	Primary Responsibilities
Project Manager	1.0	Planning, stakeholder management, risk tracking, phase gates
Lead Data Scientist	1.0	Model design, evaluation framework, hypothesis testing
Senior ML Engineer	1.0	MLOps, CI/CD, model serving, drift monitoring
Data Engineer	1.0	Pipeline development, feature store, data governance
Business Analyst	0.5	Requirements, UAT co-ordination, dashboard spec
AI Ethics & Compliance Lead	0.5	Bias audits, model cards, regulatory documentation
Total	5.0 FTE	

Table 11 — Team Structure

9. Conclusion

This project plan presents a rigorous, end-to-end blueprint for deploying a production-grade predictive analytics solution grounded in sound ML principles, ethical governance, and scalable engineering. The six-phase roadmap — spanning data engineering, model development, explainability, deployment, and continuous monitoring — is designed to deliver measurable business value within six months while laying the architectural foundation for enterprise-wide AI adoption.

The initiative is hypothesis-driven: success is defined in terms of statistically validated improvement over a seasonal naïve baseline, ensuring that claims of model efficacy are scientifically sound rather than anecdotal. The multi-source public dataset strategy demonstrates that significant value can be extracted from open data, reducing dependence on expensive proprietary data providers.

By embedding ethical review, bias auditing, and model card documentation as first-class deliverables — not afterthoughts — the project positions the organisation as a responsible AI practitioner. The MLOps framework ensures that the system remains accurate, fair, and reliable long after the initial launch, with automated drift detection and retraining pipelines sustaining model performance through evolving market conditions.

In summary, the Predictive Analytics AI Initiative represents a strategically significant investment with a projected ROI driven by inventory cost reduction, improved service levels, and the creation of a reusable ML platform. The plan is ambitious, grounded, and executable — and positions the organisation to realise the full transformative potential of artificial intelligence.

10. References & Further Reading

- [1] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *KDD 2016*.
- [2] Oreshkin, B. N., et al. (2020). N-BEATS: Neural basis expansion analysis. *ICLR 2020*.
- [3] Lim, B., et al. (2021). Temporal Fusion Transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4), 1748–1764.
- [4] Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *NeurIPS 2017*.
- [5] Mitchell, M., et al. (2019). Model cards for model reporting. *FAccT 2019*.
- [6] Sculley, D., et al. (2015). Hidden technical debt in machine learning systems. *NeurIPS 2015*.
- [7] Makridakis, S., et al. (2022). M5 accuracy competition: Results, findings, and conclusions. *International Journal of Forecasting*, 38(4).
- [8] European Parliament (2024). EU Artificial Intelligence Act. Official Journal of the European Union.
- [9] Google. (2023). Responsible AI Practices. <https://ai.google/responsibilities/responsible-ai-practices/>
- [10] Feature Store for Machine Learning (Feast). <https://feast.dev>
- [11] MLflow: An open source platform for the machine learning lifecycle. <https://mlflow.org>
- [12] Evidently AI. (2024). Open-source ML observability platform. <https://www.evidentlyai.com>